Dikshant Sharma

sdikshant0520@gmail.com (+91)7878751627 GitHub Google Scholar Website

Education

Central University of Rajasthan, Ajmer **Sep 2022 – Aug 2024**
M.Sc. - Computer Science (spl. AI & ML) *Ajmer, Rajasthan*
Modules: AI, Advanced Algorithms, Machine Learning, Prob. & Stats, NLP

University of Rajasthan **Jul 2018 - Oct 2021**
B.C.A - Bachelor of Computer Applications *Jaipur, Rajasthan*

St. Xavier Sr. Sec. School, Jaipur **May 2018**
Higher Secondary Education, CBSE (Physics, Chemistry, Maths with I.P.) *Jaipur, Rajasthan*

Experience

Research Assistant | ***Vision Language and Learning Group*** | ***IIT Jodhpur*** **May 2025 - Present**

- Advisor : ***Dr.Anand Mishra***
- **Vision-Language Models for Autonomous Driving**: Working on our group's **Bosch** research project, with additional mentorship from **Yasaswi Bharadwaj** (Research Scientist, Bosch) and **Prof. Sahana Prabhu** (IIIT Bangalore), focusing on developing efficient and reliable VLM-based approaches for domain-specific autonomous-driving applications.
- **Multimodal scene-text image retrieval**: Working on multimodal, scene-text aware image–text retrieval models for multilingual settings.
- Served as an Emergency Reviewer for **ICVGIP 2025**, contributing to the peer-review process of computer vision and image processing research submissions.
- Engage in weekly reading-group meetings focused on discussing recent research papers in ML/AI.
- Assisting in mentoring Master's and undergraduate students on vision-language research projects.

Research Assistant | Dept. of CS | *Central Univ. of Rajasthan* **Dec 2023 - July 2024**

- Advisor : ***Prof. Ajay Indian***
- Designed and implemented a lightweight convolutional neural network (CNN) for multi-class brain tumor classification, optimized to reduce model size while maintaining high classification performance.

Data Science Intern | *Mr. S. Bhatia* | *Pysliq* **Jan 2024 - Feb 2024**

- Performed comprehensive data collection, preprocessing, analysis, and interpretation to derive meaningful insights for the Number Recognition project.

Publications

- **X-STAR: Cross-lingual Scene Text-aware Image Retrieval**
  Dikshant Sharma*, Prateek, Anand Mishra*
  Submitted

- **An improved CNN model for characterizing brain tumors using deep learning**
  Ajay Indian, Gaurav Meena*,* ***Dikshant Sharma****
  Multimedia Tools and Applications(MTAP)
  *(*equal contribution)*

Projects

- **Improving IndicPhotoOCR for Multilingual Scene Text** : Working on improving the performance of IndicPhotoOCR, the lab's scene-text OCR toolkit for Indian languages, and on applying it to multilingual image captioning and word-matching tasks.
- **Uncertainty Estimation for VLMs in the Medical Domain** : Exploring uncertainty estimation for vision-language models on medical tasks, to identify when their responses are unreliable.

- **Brain Tumor Detection** : Worked on the classification of various brain tumor diseases by developing a robust deep learning model that achieved 97.24% accuracy. The model demonstrated strong capability in detecting early-stage brain tumors, contributing to improved diagnostic decision support. The project involved designing and training convolutional neural networks and applying advanced deep learning techniques to enhance reliability.
- **Stock Price Prediction** : Developed a predictive model for Microsoft stock prices using Long Short-Term Memory (LSTM) networks, leveraging historical market data from 1986 to 2022 to capture long-term temporal dependencies. Conducted analysis of multi-decade stock trends to enhance model performance and provide insights into market behavior.
- **Hotel Management System in PHP** : Developed a web-based system using PHP, HTML, and MySQL, featuring separate admin and user sections. The platform enables administrators to manage room details, perform updates, handle bookings, and process payments, ensuring smooth and efficient system operations.

Relevant Coursework

- **Indtroduction to large language Model by Mitesh M. Khapra.**
- **DeepLearning - Mitesh Khapra, SKS Iyengar IIT Ropar & IIT Madras (NPTEL)**
- **CS231N**: Deep Learning for Computer Vision from Stanford University

Technical Skills

Programming Languages: Python, SQL, PHP, C, C++
Frameworks/Libraries: PyTorch, Numpy, Sklearn, Matplotlib, Pandas
Technologies: Git, GitHub, LaTEX
Others: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Artificial Intelligence, Hadoop

Certificates & Qualifications

- **National Eligibility Test - Junior Research Fellowship(CS & Applications))** *December 2024*
- **Graduate Aptitude Test in Engineering(Computer Science Engineering)** *February 2025*
- **Two-Week Faculty Development Programme on Deep Learning and Image Processing at Central University of Rajasthan** *October 2023*
- **German Language : Two-year Diploma, University of Rajasthan** *October 2021*

Languages

- **English :** Fluent
- **German :** Intermediate
- **French :** Basic